From Natural Language to Crop Type Distribution

An agentic approach to Earth Observation

Pinaki Banerjee, Dr. Miro Miranda Lorenz, Dr. Rocco Sedona, Prof. Dr. Gabriele Cavallaro

Challenges in Earth Observation Applications

Complex Workflows :	Data retrieval, Pre-processing, Inference, Validation, Summarization
Heterogeneous Data and Tools :	Multiple sources/services, harmonization, Model specific pre-processing
Data Uncertainty :	Revise workflow for missing observation
Propagation of constraints :	Spatial, Temporal and quality requirements across processing stages
Traceability of analysis :	Provenance completeness and reproducibility

Challenges in Earth Observation Applications

- EO downstream applications are inherently complex
  - Single GeoFM solution is infeasible
  - Need multi-step analysis and domain specific tools : increases pipeline complexity !

- Code generation agents achieves 40% accuracy on EO tasks [2]

- Need dynamic solutions based on application constraints
  - **Workflow orchestration** instead of automation

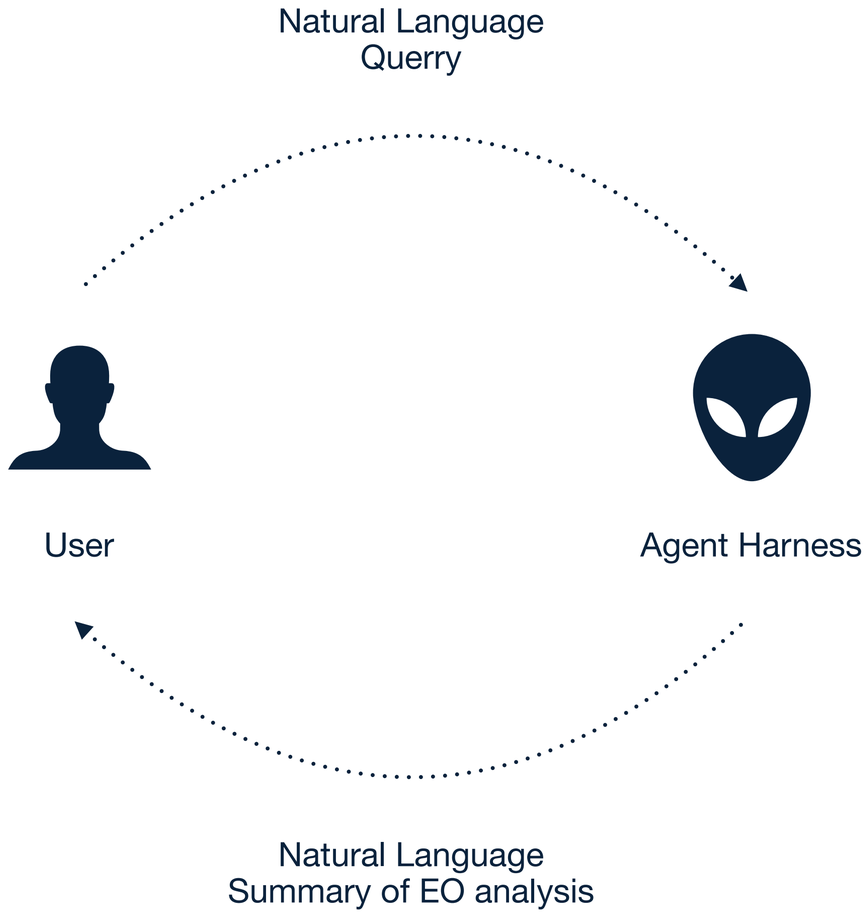
- Agentic AI offers a way to transform question to workflows which can react/adapt to intermediate results

An Agentic EO Use Case Study:

Crop Type Distribution using Terramind GeoFM

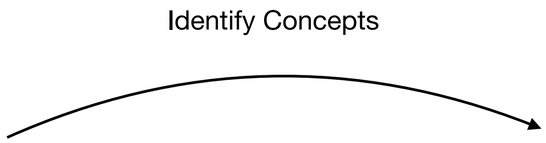
An Agentic EO Use Case Study:

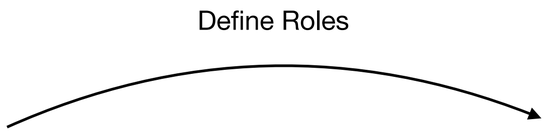
Crop Type Distribution using Terramind GeoFM



User Prompt : Breakdown and Roles

Please give me the crop-distribution for an area of 100KM2 around Dambroich in Hennef, NRW based on May to September 2017.





Prompt Element	Interpretation	Task Role
Crop distribution	Spatial distribution of crop types	**Objective:** determine “which” crops grow “where”
Dambroich, Hennef, NRW	A geographic place	**Spatial reference:** locate the study area
100 km²	Total size of the study area	**Spatial constraint:** limit its extent
May–September 2017	Observation period	**Temporal constraint**: restrict the observation period

Tool-Guided Task Decomposition and Constraint Propagation

Agent		MCP Server
Do you have some tool to calculate crop-distribution for a study area ?	◄·····►	I can give you a crop-distribution based on terramind-v1-large GeoFM but I need 3 temporal timestamps for the Sentinel-2 bands: [B01, B02, B03, B8A, B11, B12] as a AI ready dataset.
Do you have some tool for creating AI ready dataset for inference ?	◄·····►	I can prepare an AI ready Zarr dataset for you but I need a set of GeoTIFF files and the target GeoFM
Do you have a tool for downloading satellite data ?	◄·····►	I can give you any satellite data, if you provide me : • Spatial Extent • Temporal Extents • Sensors and spectral bands

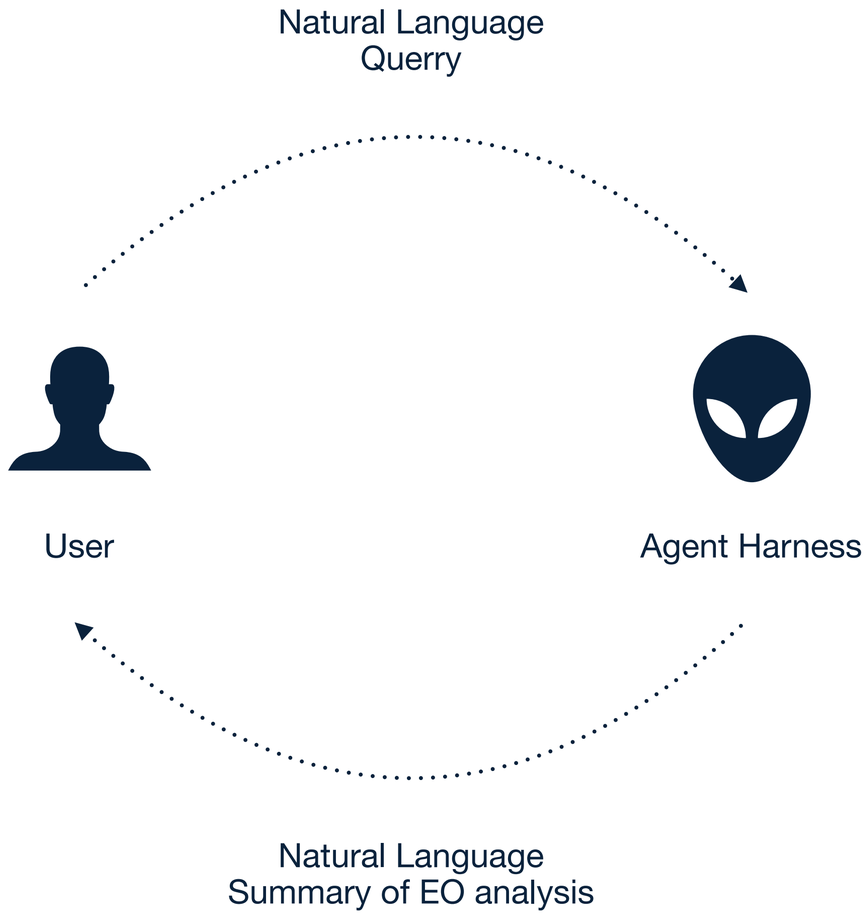
Order of Execution

Primary Goal
Objective: determine “which” crops grow “where”
Spatial reference: Dambroich, Hennef, NRW
Spatial constraint: 100 km²
Temporal constraint: May–September 2017
Tool: *mcp-eo-utils:crop_type*

Sub Goal
Objective: Create AI Ready Dataset
Target GeoFM : Terramind-v1-large
Temporal constraint: 3 timestamps : *Use Aggregation*
Input Constraint: Sentinel-2; [B01, B02, B03, B8A, B11, B12]
Tool: *mcp-eo-utils:preprocess*

Sub Goal
Objective: Download satellite data
Spatial and Temporal Constraints: *Derived from above*
Format constraint: GeoTIFF
Tool: *mcp-eo-utils:download*

System Architechture



MCP Tools



- Open-source standard [5] to enable AI applications to use external tools
- Designed to be Model-Controlled
  - ➡ LLMs can discover and invoke based on its context

MCP Tools : FastMCP

- Framework for exposing Python functions as MCP tools

- Simple, Decorator based : @tool
  - Inspects function’s signature and type annotations
  - Derives Input and Output schemas
  - Passes it to the harness which gets delivered to the LLM

```python
from fastmcp import FastMCP

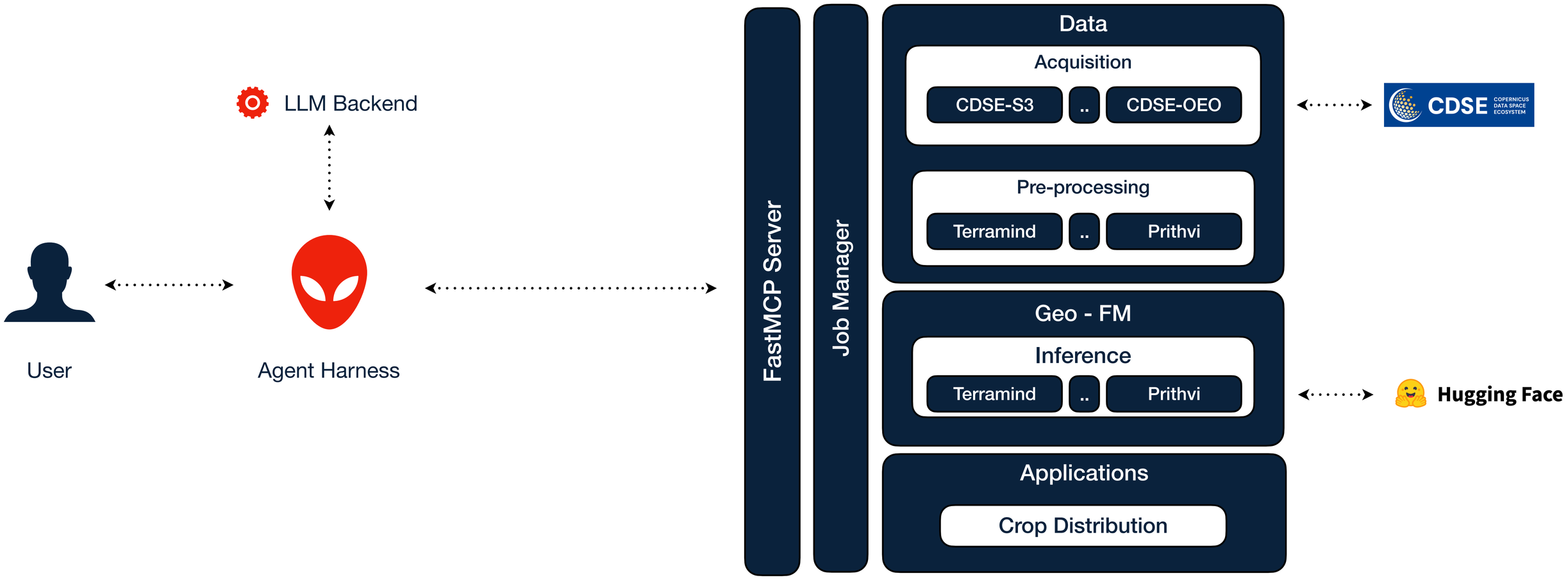
mcp = FastMCP("eo-utils", strict_input_validation=True)

def main(host: str, port: int) -> None:
   logging.info(f"Starting eo-utils MCP server on http://{host}:{port}/mcp”)
    mcp.run(transport="streamable-http", host=host, port=port)

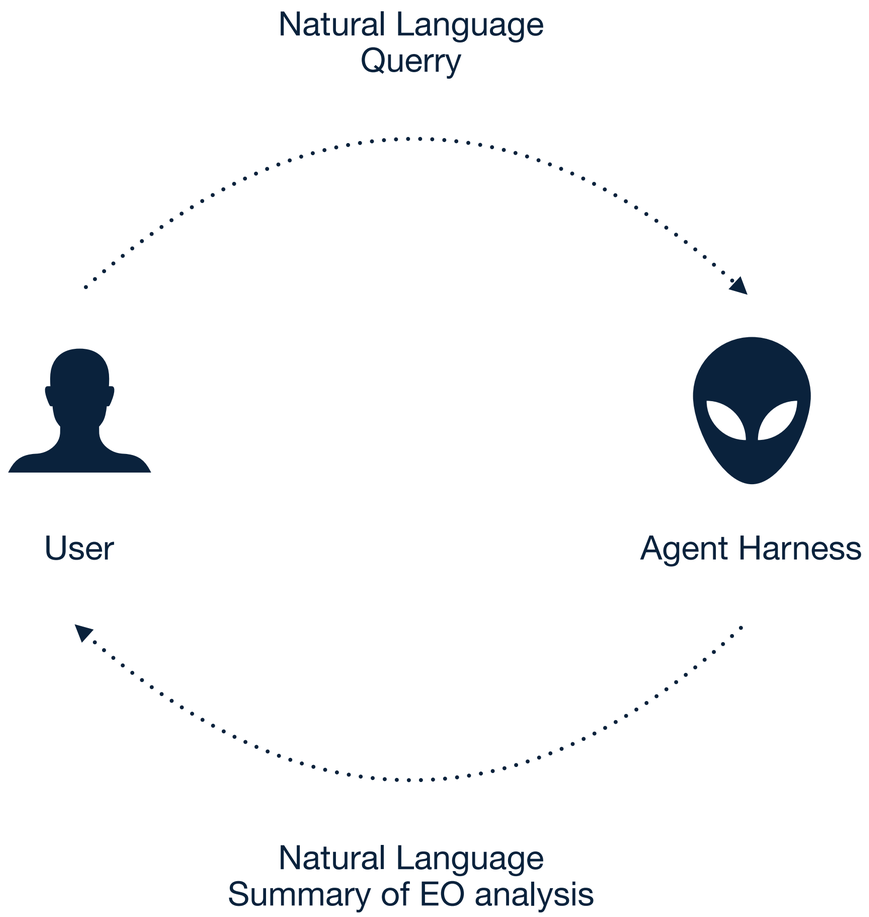
@mcp.tool()
async def download_eo_data(
    collection_id: Annotated[str, "The OpenEO collection ID"],
    spatial_extent_west: Annotated[float, "Western boundary of the spatial
extent in degrees longitude."],
      .....
    temporal_extent_start: Annotated[str, "Start date of the temporal extent
in ISO 8601 format (YYYY-MM-DD)."],
    temporal_extent_end: Annotated[str, "End date of the temporal extent in
ISO 8601 format (YYYY-MM-DD)."],
    bands: Annotated[list[str], "List of band names to download from the
collection."],
    cloud_coverage_max: Annotated[float, "Maximum cloud coverage percentage
(0-100) for filtering images."],
    aggregation: Annotated[
        Literal["year", "month", "none"],
        "Aggregation method for the downloaded data. Options: 'year',
'month', or 'none'. Use 'none' for no aggregation."
    ],
    ....,
    ctx: Context
):

.....
```

Architecture

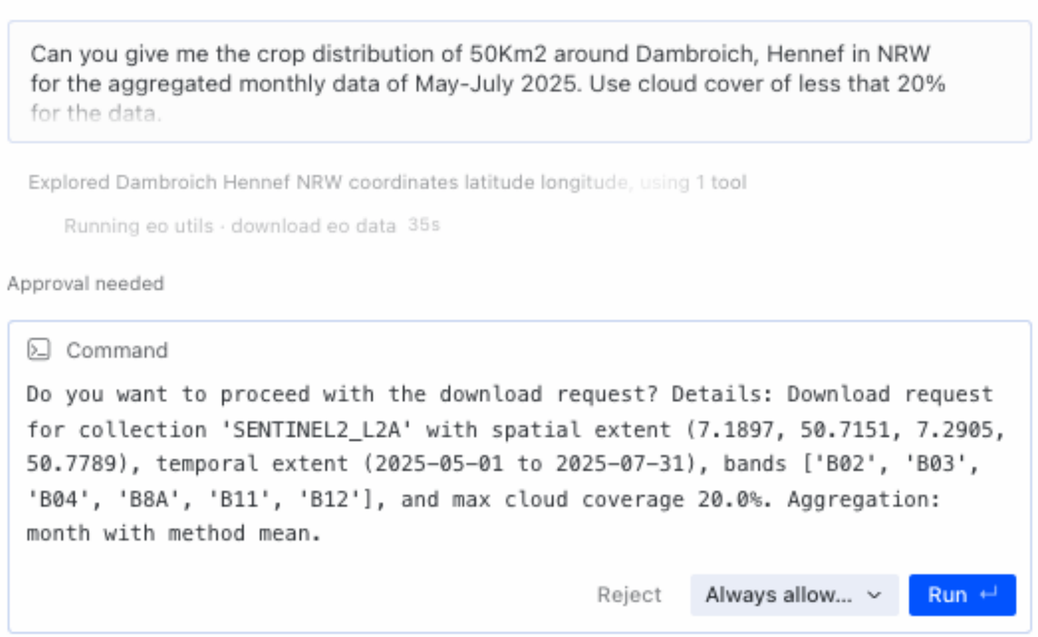


Pitfalls


Natural Language
Querry
User
Agent Harness
Natural Language
Summary of EO analysis

EO-Utils: Handle Pitfalls
Guard Rails

- Keep tool use within authorized, safe, and task-appropriate limits
- A constraint specifies a condition; a guardrail checks or enforces it
- Agents cannot respond on behalf of the user
  - Need explicit approval
- Download tool : consumes user credentials
  - Early failure detection and prevents hallucinated tool calls



EO-Utils: Handle Pitfalls
Skills

- On demand procedural knowledge for guiding and agent for a task completion
- A skill can specify:
  - **When to use it**: applicable tasks and prerequisites.
  - **How to proceed**: steps, decision criteria, and tool usage.
  - **How to validate results**: quality checks and expected outputs.
- In Hermes, skills are continuously learned
- Can be distributed and reutilized in resource constrained environments

eo-utils-crop-type General Learned

Generates crop-type distribution from eo-utils data.

Edit Archive

name	eo-utils-crop-type
trigger	Use when you need a crop-type distribution from the eo-utils MCP server.
description	Generates crop-type distribution from eo-utils data.

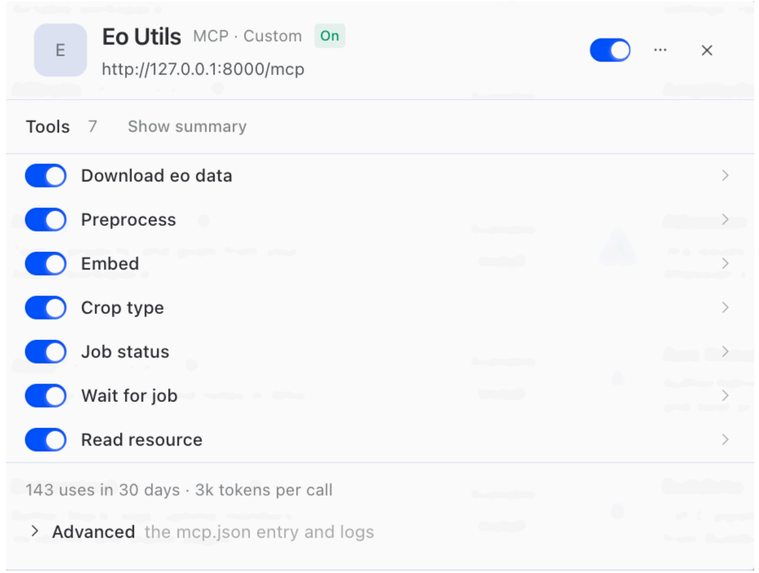
```
## Procedure
1. `mcp__eo_utils__list_resources` – confirm available tools.
2. `mcp__eo_utils__download_eo_data` with `aggregation="month"` (or `none` if you supply
multiple dates) and required bands `B02 B03 B04 B8A B11 B12`. Use collection ID
`SENTINEL2_L2A`.
3. `mcp__eo_utils__wait_for_job` for the download.
4. `mcp__eo_utils__preprocess` – target model `terramind`, modality `S2L2A`, bands list,
`data_dirs` pointing to the download output.
5. `mcp__eo_utils__wait_for_job` for preprocessing.
6. `mcp__eo_utils__crop_type` – base_model `terramind_v1_large` and the path to the
preprocessed Zarr.
7. `mcp__eo_utils__wait_for_job` for the crop-type job.
8. Read `remarks.crop_distribution_summary` from the job result or inspect `output_path` for
`stitched_mask.png`.

## Pitfalls
- Pre-processed Zarr must contain ≥ 3 temporal observations; a single-month aggregation will
fail.
- Band `B8A` is required for the crop-type model; omit it and the job errors.
- Collection ID is case-sensitive (`SENTINEL2_L2A`).
- Use the same spatial extent for download, preprocess and crop-type steps.
- Verify the job succeeded (`status":"succeeded"`) before reading the mask.
- Always display the stitched mask image (`stitched_mask.png`) to the user.

## References
- `eo-utils://embedding-notes`
- `eo-utils://preprocessing-notes`
```

EO-Utils: Handle Pitfalls
LLM context limits

- Task complexities (explicit vs implicit) : iterations and sub-goals
- Exhausted context results in
  - Hallucinations, Broken chain of thoughts, thought-loops
- Available Context Size is affected by
  - MCP Tool descriptions and outputs
  - Skills and External References
  - Progressively adds up with long chain-of-thoughts



- Limit number of tasks from the mcp-server

EO-Utils: Handle Pitfalls
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- Limit number of tasks from the mcp-server
- Separate task-to-LMM pairing based on complexities

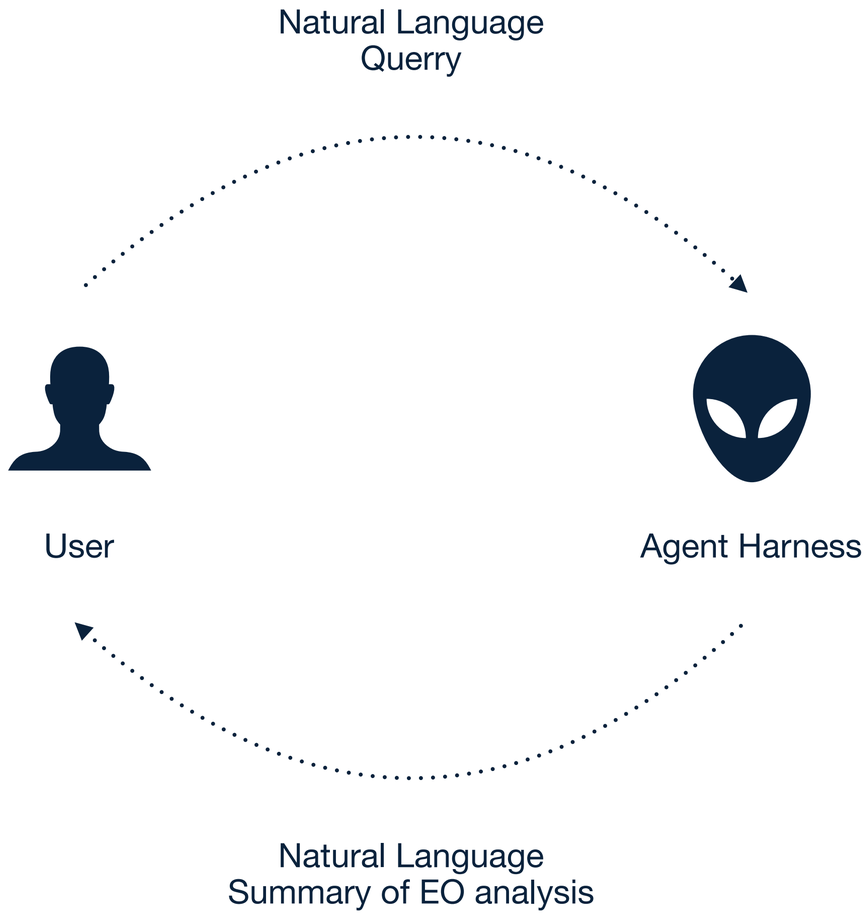
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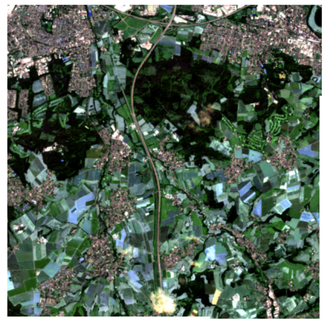
Results


Natural Language
Querry
User
Agent Harness
Natural Language
Summary of EO analysis

Results : Crop Type Distribution

Can you give me the crop distribution of 50Km2 around Dambroich, Hennef in NRW for the aggregated monthly data of May-July 2025. Use cloud cover of less that 20% for the data.

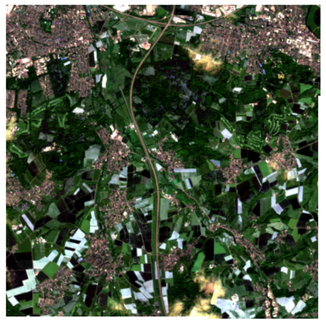
openEO_2025-06-01Z.tif

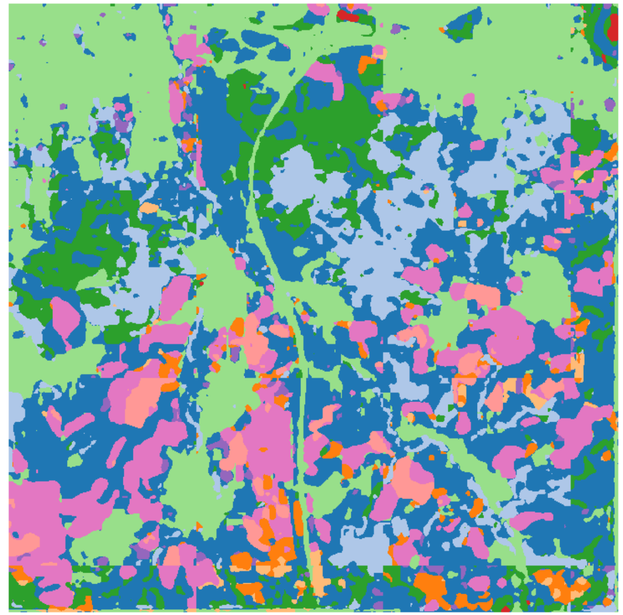


openEO_2025-07-01Z.tif



openEO_2025-05-01Z.tif





🔍 **Key Observations**

1. **Natural ecosystems** dominate (~70% combined: **Natural Vegetation → Forest → Wetlands**). → The area has **low agricultural intensity** compared to the prior (July–Sept) results.
2. **Developed/Barren Land (26.92%)** is **significantly higher** than in the summer months—likely due to **urban expansion, construction sites, or bare ground** in fall.
3. **Winter Wheat (2.51%)** appears—**expected for this season**, as it is sown in autumn.
4. **Corn (2.72%), Soybeans (0.96%), Alfalfa (1.18%)** are minimal **compared to summer results**, suggesting **harvest or dormancy** in these crops.
5. **Open Water (0.14%), Cotton, Fallow, and Sorghum (0.0%)** are nearly absent.

Results : Embedding Generation

mcp-eo-utils Hi, use mcp-eo-utils and please give me the terramind embeddings for an area of 100KM2 around Paris for May to September 2017 with maximum cloud coverage of 20%. Use Sentinel2 data and RGB bands only.

Pipeline complete. Summary:

1. Download (job `25066ab654b3`) — succeeded

- Collection: SENTINEL2_L2A, bands B02/B03/B04 (RGB)
- Area: ~100km² around Paris (2.2838°E–2.4206°E, 48.8116°N–48.9016°N)
- Temporal: 2017-05-01 to 2017-09-30, cloud coverage ≤20%
- Preview: `output/preview.png` at `/Users/pinakibanerjee/Workspace/codebase/EO_UTILS/eo-utils/experiments/agentic/jobs/25066ab654b3/output/preview.png`

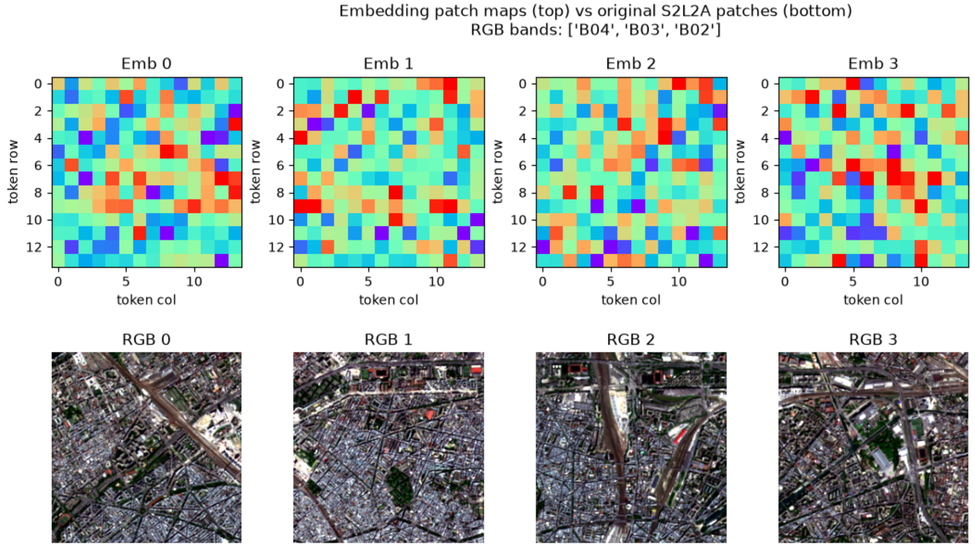
2. Preprocess (job `efeb8ce3dee4`) — succeeded

- Target model: terramind, modality S2L2A (BLUE/GREEN/RED)
- Zarr output: `.../efeb8ce3dee4/output/preprocessed.zarr`

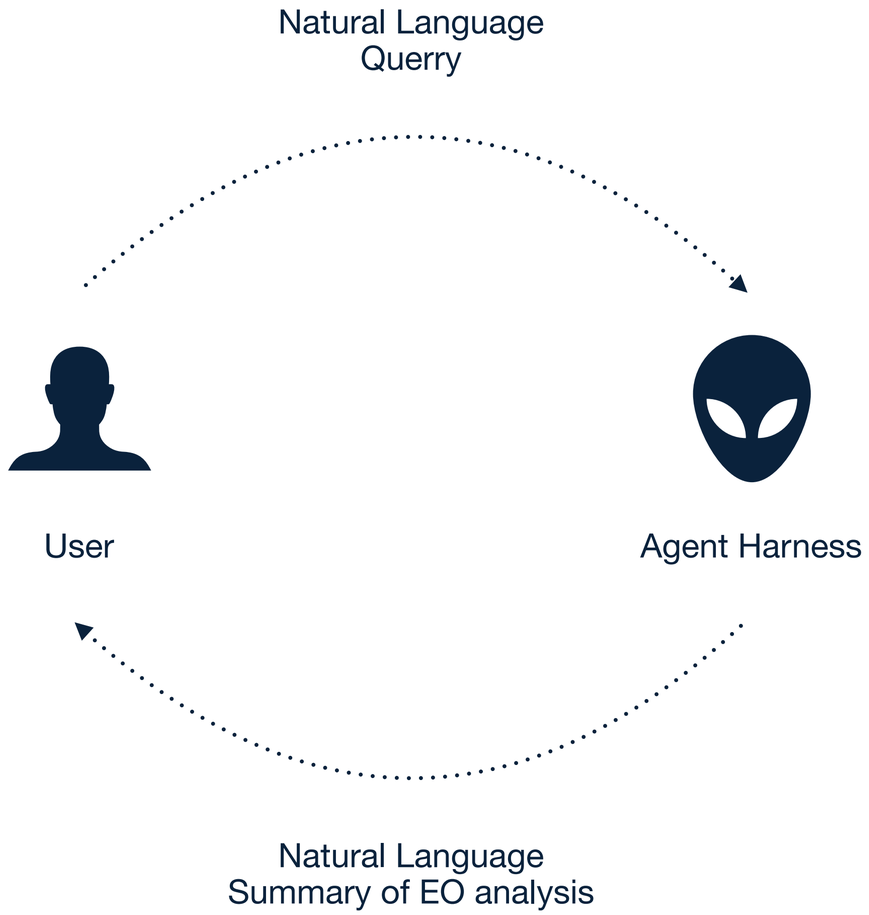
3. Embed (job `fafbafbbf0fb`) — succeeded

- Model: terramind_v1_large
- Output (Zarr embeddings array): `/Users/pinakibanerjee/Workspace/codebase/EO_UTILS/eo-utils/experiments/agentic/jobs/fafbafbbf0fb/output/`

The embeddings are stored as a Zarr array at that path (`embeddings/` group, chunked), ready for downstream use.



Evaluation and Next steps



Evaluation

- Our experiments demonstrate promising tool orchestration, but also failures
- Scientific value of our pipelines
  - Analytical validity and task completion
  - Acceptable cost :
    - Operational
    - Failures

- Related work :
  - ReAct [1] framework based evaluation strategies
  - Earth-Bench [3]: Represents agentic workflows as Partially Observable Markov Decision Processes; provides curated dataset
  - UnivEARTH [2]: 408 yes/no questions spanning 7 diverse topics and 15 main sources
  - [4] provides a comprehensive survey of different evaluation frameworks in the current SOA

Next Steps

- Make it accessible :
  - Public Github repository for local MCP server setup
  - Host a public accessible MCP server
    - User authentication
    - User data privacy

- Quality control of MCP tools
  - Evaluate the agentic system w.r.t established benchmarks
  - Evaluate the individual tools w.r.t LLM backends with different context sizes
  - Evaluate tool APIs w.r.t EO task complexities
    - process-graphs, combining multiple predecessor outputs for a downstream task
  - Create a selection/recommendation for tools for different user setups

- Expand with other GeoFM support and more downstream applications

Can we already use this ?

- Local server setup is already functional
  - https://gitlab.jsc.fz-juelich.de/sdlrs/eo-utils/-/tree/pb/agentic?ref_type=heads
- Install the project and run the following

  python -m eo_utils.agentic

  - (Or simply start the docker container)



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  python -m eo_utils.agentic

  - (Or simply start the docker container)
- Hook it up with any agentic harness
  - For Hermes, just add it to the config file
- That's it !

```yaml
mcp_servers:
  eo-utils:
    url: http://127.0.0.1:8000/mcp
    timeout: 300
    elicitation:
      enabled: true
      timeout: 300
    tools:
      include:
        - download_eo_data
        - preprocess
        - embed
        - crop_type
        - job_status
        - wait_for_job
        - read_resource
      resources: false
      prompts: false
```

References

1. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K. and Cao, Y., 2022. ***ReAct: Synergizing reasoning and acting in language models.*** arXiv preprint arXiv:2210.03629.

2. Kao, C.H., Zhao, W., Lam, C., Umap, A., Revankar, S., Speas, S., Bhagat, S., Datta, R., Phoo, C.P., Mall, U. and Vondrick, C., 2026, July. ***Towards llm agents for earth observation.*** In Findings of the Association for Computational Linguistics: ACL 2026 (pp. 2597-2611).

3. Feng, P., Lv, Z., Ye, J., Wang, X., Huo, X., Yu, J., Xu, W., Zhang, W., Bai, L., He, C. and Li, W., 2026, April. ***Earth-agent: Unlocking the full landscape of earth observation with agents.*** In International Conference on Learning Representations (Vol. 2026, pp. 55700-55764).

4. Talemi, N.A., Boone, J. and Afghah, F., 2026. ***Agentic ai in remote sensing: Foundations, taxonomy, and emerging systems***. arXiv preprint arXiv:2601.01891.

5. ***Model Context Protocol (2026) Specification.*** Version 2026-07-28. Available at: https://modelcontextprotocol.io/specification/2026-07-28 (Accessed: 12 October 2026).

Thank You !